

1. Administrative & Study Identification

**Brief Study Title:** Study of Novel Immunomodulators as Monotherapy and in Combination With Anticancer Agents in Participants With Advanced Hepatobiliary Cancer **Official Title:** A Phase II, Open-Label, Multi-Drug, Multi-Center, Master Protocol to Evaluate the Efficacy and Safety of Novel Immunomodulators as Monotherapy and in Combination With Anticancer Agents in Participants With Advanced Hepatobiliary Cancer **Short Title/Acronym:** GEMINI-Hepatobiliary **Protocol Number:** D7987C00001 **NCT Number:** NCT05775159 **Sponsor/Company:** AstraZeneca **Investigational Product:** Novel immunomodulators (Volrustomig and Rilvegostomig) **Phase of Development:** Phase II **Trial Status:** Recruiting **Medical Monitor:** AstraZeneca Clinical Operations Lead

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2. Study Synopsis & Rationale

**2.1 Study Objective Primary Objective:** To evaluate the preliminary efficacy, safety, and tolerability of novel immunomodulators alone and in combination with other anticancer drugs in participants with specified advanced solid tumors (hepatocellular carcinoma [HCC] and biliary tract cancer [BTC]). **Secondary Objectives:** To evaluate duration of response (DoR), disease control rate (DCR), progression-free survival (PFS), overall survival (OS), pharmacokinetics, and immunogenicity.

**2.2 Background & Rationale** Although targeted therapies and immuno-oncology (IO)-based regimens have improved overall survival in advanced hepatobiliary carcinomas, long-term survival rates remain unsatisfactory. This GEMINI-Hepatobiliary master protocol aims to assess if novel immunomodulators (Volrustomig or Rilvegostomig) as monotherapy (MONO) and/or in combination with anticancer agents (COMBO) can improve outcomes in participants with advanced hepatobiliary cancer (e.g., HCC, BTC, etc.).

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3. Study Design & Methodology

**3.1 Design Framework Study Type:** Interventional (Master Protocol / Platform Design) **Control Type:** Uncontrolled **Blinding:** Open-label **Randomization:** Substudy-specific allocation (Modular design with independent substudies)

**3.2 Planned Enrollment & Duration Target \$N\$:** 294 evaluable participants. - *Substudy 1:* Approximately 200 evaluable participants with advanced HCC. - *Substudy 2:* Approximately 90 evaluable participants with advanced BTC. **Number of Sites:**

**4. Subject Selection (Eligibility Criteria)**

**4.1 Inclusion Criteria** 1. Age  $\geq$  18 years at the time of signing the ICF. 2. Provision of a signed and dated written ICF. 3. Confirmed locally advanced or metastatic solid tumor specified in substudy based on histopathology. (For Substudy 2: advanced BTC who have not received previous treatment for advanced/metastatic disease). 4. Adequate organ and bone marrow function. 5. At least 1 measurable not previously irradiated lesion per RECIST 1.1. 6. Life expectancy of at least 12 weeks at the time of screening. 7. Willing and able to provide an adequate tumor sample.

**4.2 Exclusion Criteria** 1. History of allogeneic organ transplantation. 2. Active or prior documented autoimmune or inflammatory disorders, and active primary immunodeficiency. 3. Uncontrolled intercurrent illness. 4. History of another primary malignancy, leptomeningeal carcinomatosis. 5. Active infection, brain metastases or spinal cord compression. 6. Participants co-infected with HBV and hepatitis D virus (HDV). 7. Previous treatment in the present study. 8. For substudy 1, history of hepatic encephalopathy within 12 months prior to treatment allocation.

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**5. Intervention & Treatment Plan**

**5.1 Investigational Regimen Dosage/Frequency:** - *Substudy 1 (HCC):* Volrustomig and Rilvegostomig evaluated as monotherapy and/or in combination with other anticancer drugs. Volrustomig typically 15 mg/kg IV on Day 1 of each 21-day cycle, or standard combination TKI doses (e.g., Lenvatinib). - *Substudy 2 (BTC):* Rilvegostomig or Volrustomig plus gemcitabine and cisplatin standard dosing. **Route of Administration:** Intravenous (IV) for mAbs and chemotherapy (cisplatin/gemcitabine); Oral (PO) for TKIs. **Duration of Treatment:** Until disease progression per RECIST 1.1 or unacceptable toxicity develops.

**5.2 Concomitant & Prohibited Therapy Permitted:** Standard supportive care; specific standard-of-care anticancer agents according to the assigned substudy. **Prohibited:** Concurrent investigational drugs or unauthorized systemic anti-cancer therapies.

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**6. Study Timeline & Visit Schedule**

| Visit ID | Window (Days) | Key Activities/Procedures | | :---  
| :--- | :--- | | **Screening** | Day -28 to 0 | Informed Consent,

Tumor Biopsy/Histopathology, Baseline Imaging (RECIST 1.1), Safety Labs | | **Baseline** | Day 0 | Substudy Allocation, First Dose of Therapy | | **Treatment Cycles** | Q3W (21 Days) | IV Infusions, Dispense Oral Meds, Safety Labs, PK/ADA Samples, Efficacy Assessment (Scans) | | **End of Study**| Variable | End of Treatment Visit upon Progression/Toxicity, Survival Follow-up |

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## 7. Outcome Measures (Endpoints)

**7.1 Primary Outcomes** 1. **Objective response rate (ORR)**: Proportion of participants who have a confirmed CR (complete response) or confirmed PR (partial response), determined by the Investigator at local site per RECIST 1.1 (For HCC sub-study 1). 2.

**Progression free survival (PFS)**: Time from the start of study intervention until progression per RECIST 1.1 as assessed by the Investigator at the local site or death due to any cause in the absence of progression, whichever occurs first (For BTC sub-study 2). 3. **Number of participants with adverse events/serious adverse events**: Including abnormal clinical observations, abnormal Electrocardiogram (ECG) parameters, abnormal laboratory assessments and abnormal vital signs that changed from baseline.

**7.2 Secondary Outcomes** 1. Duration of Response (DOR). 2. Disease Control Rate (DCR). 3. Progression free survival (PFS).

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## 8. Safety & Adverse Event Reporting

**Adverse Event (AE) Monitoring**: Continuous monitoring of abnormal clinical observations, ECG parameters, laboratory assessments, and vital signs compared to baseline. **Serious Adverse Events (SAEs)**: Standard 24-hour reporting timeline for severe and unexpected events. **Data Monitoring Committee (DMC)**: Internal safety review committee to oversee platform-level toxicities.

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## 9. Statistical Analysis Plan (SAP) Summary

**Analysis Populations**: Intent-to-Treat (ITT) for efficacy evaluation; Safety Set for AE/SAE and tolerability. **Sample Size Justification**:  $N=294$ ; Sample sizes were selected to provide adequate precision for the primary endpoints. **Handling of Missing Data**: Standard right-censoring methods for time-to-event endpoints (PFS/OS).

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## 10. Quality Assurance & Regulatory Compliance

**GCP Compliance**: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

**Monitoring Plan**: Regular onsite and remote monitoring for master

protocol adherence across global sites. **Audit Trail:** Validated eCRF capturing RECIST 1.1 measurements, lab parameters, and investigational product accountability.

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## 11. Study Identifiers & Governance

**Primary ID:** D7987C00001 **Registry Number:** NCT05775159 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** GEMINI-Hepatobiliary **Official Regulatory Title:** A Phase II, Open-Label, Multi-Drug, Multi-Center, Master Protocol to Evaluate the Efficacy and Safety of Novel Immunomodulators as Monotherapy and in Combination With Anticancer Agents in Participants With Advanced Hepatobiliary Cancer

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## 12. Clinical Framework (Oncology Specific)

**Disease Areas:** Hepatocellular Carcinoma, Biliary Tract Cancer **Diagnosis Threshold:** Histologically confirmed locally advanced or metastatic solid tumor with  $\geq 1$  measurable lesion per RECIST 1.1 **Medical Methodology:** IO-based targeted therapy (Novel immunomodulators) +/- standard of care chemotherapy (gemcitabine and cisplatin) / TKIs **Optimization Target:** Objective tumor response (ORR) and delayed disease progression (PFS)

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## 13. Study Procedures & Methodology

**Management Pathway:** Master protocol utilizing separate, independent substudies (Substudy 1 for HCC, Substudy 2 for BTC) **Education Protocol (HCP):** Investigator training on RECIST v1.1 evaluation and managing IO-related immune-mediated adverse events. **Education Protocol (Patient):** Education on treatment adherence and recognizing immune-related adverse effects. **Quality Audit Mechanism:** Centralized radiological review processes and strict ADA / PK sample handling tracking.

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## 14. Observation & Follow-Up Schedule

**Total Duration:** Estimated 3-4 years (Follow-up until death or data cutoff) **Onsite Visit Cadence:** Every 21 days corresponding with the day 1 of each treatment cycle **Remote Monitoring Frequency:** Continuous safety and survival follow-up post-treatment discontinuation **Checklist Review Interval:** Imaging assessments per protocol defined intervals

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## 15. Sign-off & Approval

| Role                     | Name   | Signature | Date          |  | :--- | :--- | :--- | :--- |
|--------------------------|--------|-----------|---------------|--|------|------|------|------|
| Principal Investigator   | [Name] | _____     | [DD-MMM-YYYY] |  |      |      |      |      |
| Clinical Operations Lead | [Name] | _____     | [DD-MMM-YYYY] |  |      |      |      |      |
| Statistician             | [Name] | _____     | [DD-MMM-YYYY] |  |      |      |      |      |